A Capstone Project Report

on

Samastha Abroad Platform

Submitted in partial fulfillment for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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CERTIFICATE OF COMPLETION

This is to certify that the work entitled, **“Samastha Abroad Platform”** is the bonafide work of **Mounika Sri Lakshmi Durga Bandi (N190263)** carried out under my guidance and supervision for 4th year project of Bachelor of Technology in the department of Computer Science and Engineering under RGUKT IIIT Nuzvid. This work is done during the academic session July 2024 – April 2025, under our guidance.

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CERTIFICATE OF EXAMINATION

This is to certify that the work entitled, **"Samastha Abroad Platform"** is the bonafide work of **Mounika Sri Lakshmi Durga Bandi (N190263)** and hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in 4th year of **Bachelor of Technology** for which it has been submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn, as recorded in this thesis. It only signifies the acceptance of this thesis for the purpose for which it has been submitted.

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DECLARATION

I, **Mounika Sri Lakshmi Durga Bandi (N190263)** hereby declare that the project report entitled **“Samastha Abroad Platform”** done by me under the guidance of **Mr. K. Sravan Kumar**, Assistant Professor, is submitted for the partial fulfillment of the award of degree of Bachelor of Technology in Computer Science and Engineering during the academic session July 2024 - April 2025 at RGUKT-Nuzvid.

I also declare that this project is a result of my own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

Date: 25-04-2025 B.M.S.Lakshmi Durga (N190263)

Place: Nuzvid

ACKNOWLEDGEMENT

I would like to express my profound gratitude and deep regards to my guide **Mr.K.Sravan Kumar** for his exemplary guidance, monitoring, and constant encouragement throughout the B.Tech course. I shall always cherish the time spent with him during the course of this work due to the invaluable knowledge gained in the field of software development.

I am extremely grateful for the confidence bestowed upon me and for entrusting me with my project entitled **“Samastha Abroad Platform”**.

I express my gratitude to **Mrs. S. Bhavani** (HOD of CSE) and other faculty members for being a source of inspiration and constant encouragement, which helped me in completing the project successfully.

Finally, yet importantly, I would like to express my heartfelt thanks to my beloved God and parents for their blessings, and my friends for their help and wishes for the successful completion of this project.

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0.1 Abstract

The Samastha Abroad Authentication System is a secure and user-centric digital access solution designed for an educational and career guidance platform serving students. This system addresses the need for a flexible and role-aware authentication process by implementing a dual-login mechanism that supports both email and phone number verification methods [6].

Once credentials are validated, the platform dispatches a One-Time Password (OTP) for user verification, enhancing security while maintaining ease of use. The system integrates role-based redirection logic, guiding users to appropriate services such as AI Mentor, Career Accelerator, Soft SkillX, or Abroad Guidance, based on their assigned permissions [7]. This not only ensures relevance in content delivery but also optimizes user engagement.

To improve reliability and user satisfaction, the system includes fallback options for unregistered inputs, incorrect OTPs, and retry paths that offer alternative login routes [9]. These features contribute to a smooth and intuitive user experience, essential for a dynamic learning support platform.

By combining robust authentication practices with intelligent user routing, the system provides a secure, scalable, and personalized access model for students accessing critical educational tools [10].

0.2 Introduction

0.2.1 Motivation for the Work

The Samastha Abroad Platform offers tailored educational and career guidance through integrated services including AI Mentors, Career Accelerator, Soft SkillX, and Abroad Guidance [5]. The platform features a secure OTP-based authentication system supporting both email and phone login, with a 6-digit time-sensitive OTP delivered via email (SendGrid/AWS SES) or SMS/call (Twilio/AWS SNS) [6]. The system validates credentials, checks for proper email/phone formats, and logs failed attempts for security monitoring [9]. Upon successful authentication, users are routed based on their role (Super Admin, Admin, or Student) and subscribed services [7], with fallback mechanisms like OTP resend and login method switching available for improved user experience [6].

Built on React with TypeScript [1], Node.js, and MySQL, the platform ensures secure yet convenient access while dynamically redirecting users to appropriate interfaces. Admins access organization-specific dashboards [7], while students are directed either to individual service pages or a consolidated dashboard with progress tracking. The system implements security measures like temporary account locks after repeated OTP failures, balancing robust protection with user-friendly access to educational resources [10].

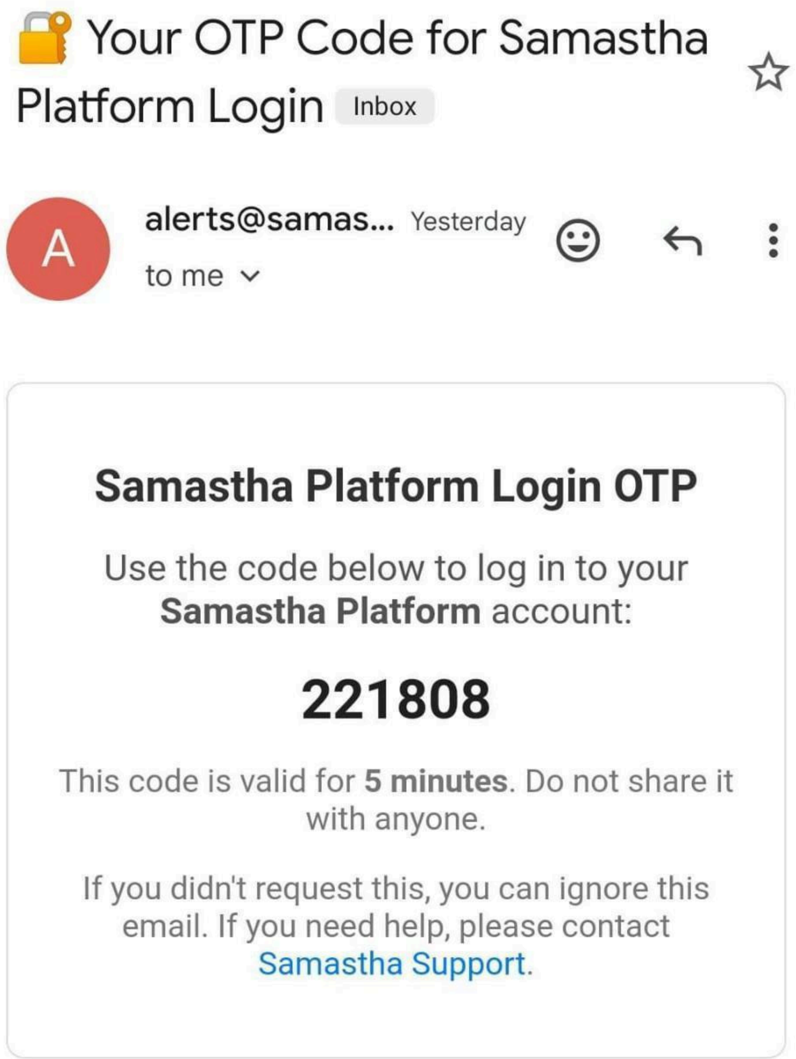

Figure 2.1: User Authentication via OTP

0.2.2 Real-world Applications

The platform's frontend architecture is built upon React's component-based model, which enables the encapsulation of authentication workflows, role-specific interfaces, and analytics dashboards into discrete, reusable units [1]. React's hooks-based state management system efficiently handles the dynamic data flows required for real-time OTP validation, role-based UI rendering, and subscription status updates, while maintaining strict separation between presentation and business logic layers [1]. The implementation leverages React's concurrent rendering features to ensure smooth interface transitions during context switches—such as when users move from authentication screens to their personalized dashboards—without compromising the cryptographic operations happening in the background [1][6]. TanStack Router's advanced routing capabilities [2] support this complex navigation by implementing permission-aware route guards that dynamically adjust available paths based on the user's RBAC profile [7], with lazy-loaded route components that align with the platform's code-splitting strategy for optimal performance. The router's integrated data loading patterns enable seamless prefetching of service entitlements and progress tracking metrics while maintaining the security boundaries established by the backend's token verification system [8].

For UI construction, Shadcn UI's accessible component library [3] provides the foundational building blocks for secure form inputs, authentication modals, and context-sensitive security prompts, all designed to WCAG standards while accommodating the platform's custom security requirements. These components are enhanced with Tailwind CSS's utility-first styling system [4], which enables rapid iteration of responsive layouts that adapt to both organizational branding guidelines and the platform's progressive disclosure patterns for complex features [10]. Lucide's icon set [5] plays a critical role in the interface's visual language, providing clear, scalable indicators for security states (such as multi-factor authentication status) and service categories while maintaining pixel-perfect clarity across the platform's mobile and desktop experiences. The styling system's deterministic class names work in tandem with React's component model to prevent CSS leakage between security-critical interface zones, an essential consideration given the platform's need to isolate high-privilege administrative controls from standard user flows [7]. This meticulous frontend architecture demonstrates how modern web technologies can satisfy Bruce Schneier's cryptographic implementation principles [9] while delivering on Jakob Nielsen's usability heuristics [10], particularly through its balanced approach to security visibility and user control.

Related Works / Existing Works

0.3 Few Related Works and their Limitations

The features described in the Samastha platform reflect best practices and patterns observed in several established educational and career guidance systems:

Existing Approaches

EdTech Platforms: These platforms incorporate user authentication through OTP, dynamic redirection based on user subscriptions or roles, and offer personalized dashboards tailored to each learner's journey.
CRM and Student Portals: Widely used in academic institutions, these systems provide role-based access control, modular service management, and personalized content delivery for students and educators.
Global Study Abroad Services: These services facilitate secure logins, profile-driven navigation, and provide comprehensive tools for mentorship, university matching, and application guidance tailored to international education seekers.

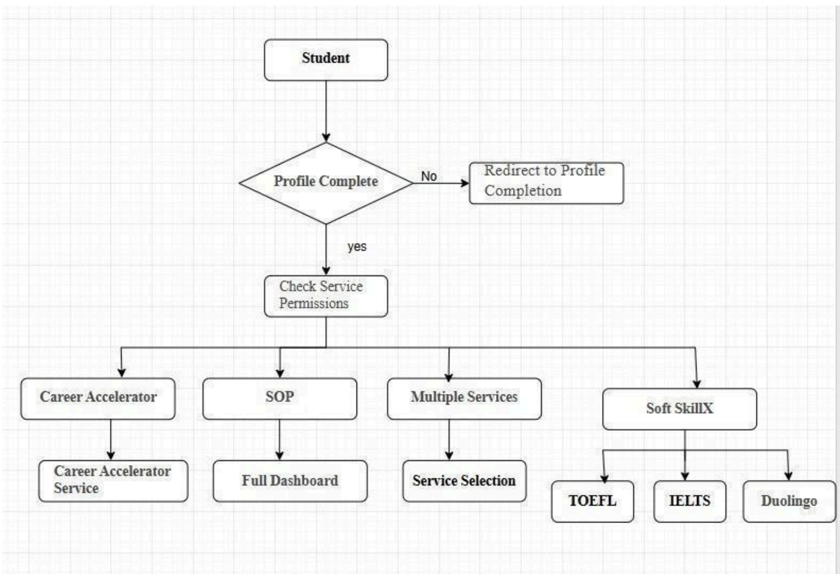

Figure 3.1: Student Flow

0.3.1 Limitations of Existing Works

Samastha addresses key limitations in existing educational platforms through an integrated and user-centric design. Unlike traditional fragmented systems that require users to manage multiple platforms for career mentoring, skill development, and study-abroad services, Samastha unifies these offerings under a single login and dashboard [1]. While other platforms provide basic personalization, Samastha dynamically redirects users based on their subscribed services, ensuring a customized interface [2]. Many systems also lack flexible login options, relying only on email or social accounts, whereas Samastha supports both email and phone-based OTP verification for broader accessibility [3]. Additionally, conventional role-based systems often have rigid permissions, but Samastha implements intelligent redirection logic to guide users to relevant dashboards based on their active services [4]. These features make Samastha a more seamless, secure, and adaptive solution for modern educational needs.

0.3.2 Research Gaps Identified

Current educational platforms exhibit several research gaps that need to be addressed. First, there is a lack of unified platforms that integrate career mentorship, skill development, and study-abroad guidance into a single cohesive experience, as most systems offer these services in isolation [1]. Second, while role-based access exists in many platforms, few implement dynamic redirection based on a user's specific combination of subscribed services, resulting in static dashboards that fail to adapt to evolving user needs [2]. Third, despite the global nature of modern education, comprehensive multi-channel OTP-based authentication supporting both email and phone login with international number recognition remains underdeveloped [3]. Finally, most platforms lack context-aware login recovery features, missing opportunities to provide adaptive support during authentication failures, such as suggesting alternative login methods or offering intuitive guidance [4]. These gaps highlight the need for more integrated, adaptive, and user-friendly educational platforms.

0.4 Proposed Method

0.4.1 Flowchart of the Proposed Approach

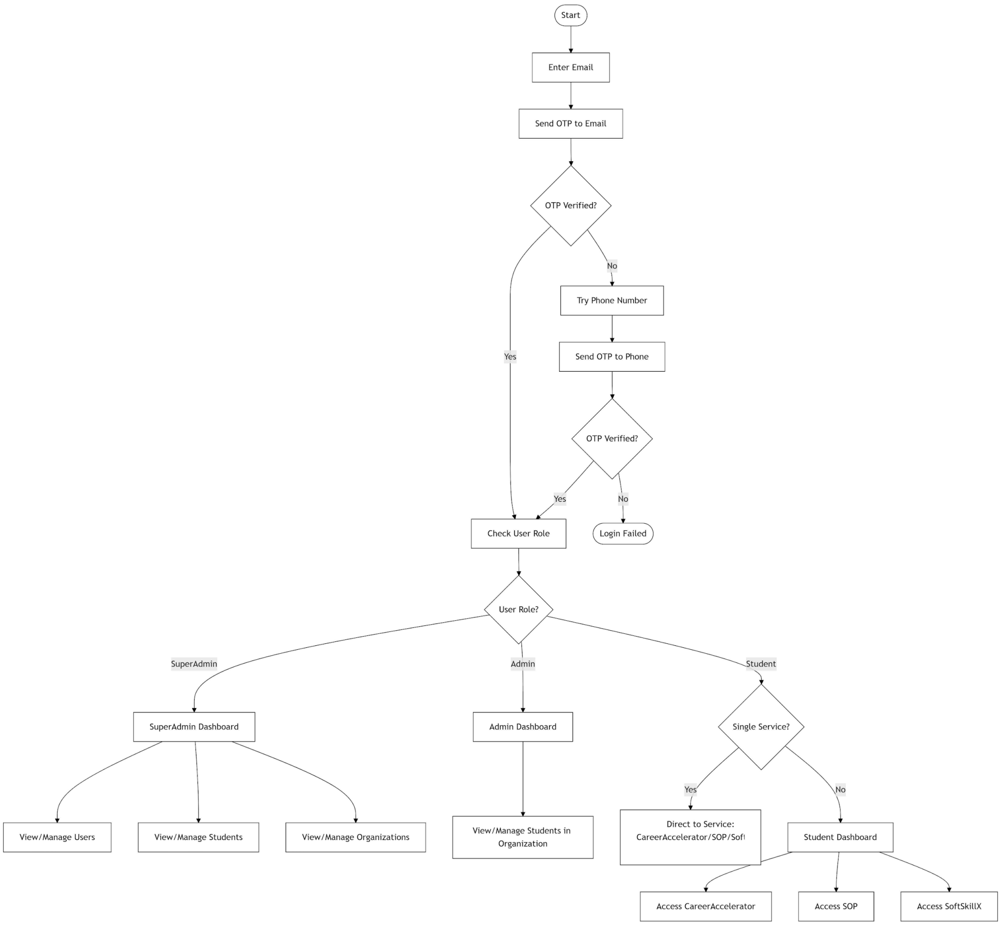

Figure 4.1: System Flowchart

0.4.2 Explanation of Each Component in the Flowchart

1. **Initiates the user login process [9]**

   The system begins by verifying user credentials through a secure authentication mechanism while fetching profile data, test history, and progress records.

2. **Collects the user's email for OTP verification [5]**

   The system collects and validates the email format (user@domain.com), rejecting invalid entries with an error prompt.

3. **Delivers a time-sensitive one-time password [6]**

   A 6-digit OTP (expiring in 5–10 minutes) is generated and delivered via email services (SendGrid/AWS SES) with attempt logging for security audits.

4. **Validates email access through OTP [6]**

   The system compares the user-entered OTP with the generated code, proceeding to role detection if matched or triggering phone fallback if failed.

5. **Activates fallback phone authentication [6]**

   When email OTP fails, the system prompts the user to enter their registered phone number for alternative verification.

6. **Sends SMS/call OTP verification [6]**

   The system delivers an OTP via SMS/call using gateways (Twilio/AWS SNS) while maintaining equivalent security protocols to email OTP.

7. **Confirms phone number ownership [9]**

   Successful verification advances to role detection while repeated failures (3–5 attempts) trigger account locking.

8. **Determines user permissions [7]**

   The system queries the database to identify the user's role (superadmin/admin/student) and routes them to the appropriate dashboard.

9. **Provides Admin Dashboard access [7]**

   Admins can view/manage students only within their assigned organization without global user/org management privileges.

10. **Checks student service access [10]**

    The system verifies subscribed services, redirecting to a specific service for single access or to the Student Dashboard for multiple services.

11. **Displays Student Dashboard [10]**

Students see all accessible services (CareerAccelerator/SOP/SoftSkillX) with consolidated progress tracking in one interface.

12. **Implements UI components for seamless navigation [3]**

    Uses Shadcn/UI's pre-built React components (e.g., modals, forms) for consistent and accessible design.

13. **Applies utility-first CSS for responsive layouts [4]**

    Leverages Tailwind CSS to ensure mobile-friendly interfaces and adaptive styling.

14. **Manages client-side routing efficiently [2]**

    Employs TanStack Router for dynamic route handling, enabling smooth transitions between dashboards.

15. **Logs authentication events for auditing [8]**

    Records OTP attempts, failures, and role-based access changes in a Node.js backend for security reviews.

16. **Encrypts sensitive data transmission [9]**

    Uses AES-256 encryption (via libraries like OpenSSL) for OTPs and session tokens during transit.

17. **Optimizes UX with progressive disclosure [10]**

    Follows Nielsen's usability principles to minimize cognitive load (e.g., step-by-step OTP input).

18. **Renders dynamic icons for visual feedback [5]**

    Integrates Lucide icons to indicate success/failure states during authentication.

19. **Handles session persistence [1]**

    Utilizes React state management (Context API) to maintain user sessions post-login.

20. **Supports role-based API access [7]**

    Restricts backend endpoints via NIST RBAC standards, ensuring admins/students access only permitted data.

0.4.2 Algorithm

```
BEGIN Login_Process
    INPUT email
    SEND OTP to email
    IF OTP_verified THEN
        role = GET_USER_ROLE()
        IF role == "SuperAdmin" THEN
            REDIRECT_TO SuperAdmin_Dashboard
        ELSE IF role == "Admin" THEN
            REDIRECT_TO Admin_Dashboard
        ELSE IF role == "Student" THEN IF
            has_single_service  THEN
                REDIRECT_TO service (CareerAccelerator/SOP/SoftSkillX)
            ELSE
                REDIRECT_TO Student_Dashboard
            END IF
        END
    IF ELSE
        INPUT phone_number
        SEND OTP to phone
        IF OTP_verified THEN
            REPEAT role_check (as above) ELSE
            DISPLAY "Login Failed" END
        IF
    END IF
END Login_Process
```

0.5 Experimental Setup

0.5.1 Technologies and Libraries Used

- **Core Technologies:** Node.js for backend services [8], React with TypeScript for frontend interface [1], MySQL for database management, TanStack Router for Routing [2], Zustand for State Management.
- **Libraries:** Axios is a promise-based HTTP client for making API requests in JavaScript and TypeScript applications. Shadcn UI is a set of accessible and customizable UI components built with Tailwind CSS and Radix UI [3][4]. Icons are integrated via Lucide [5].

0.5.2 System Requirements and hardware Requirements:

- Processor: Intel Core i5 or equivalent
- RAM: 8GB minimum
- Storage: 500GB
- Stable Internet Connection

Software Requirements:

- Operating System: Windows
- Node.js 14 or higher
- Modern web browser

0.5.3 Results

Figure 5.1: Login Page

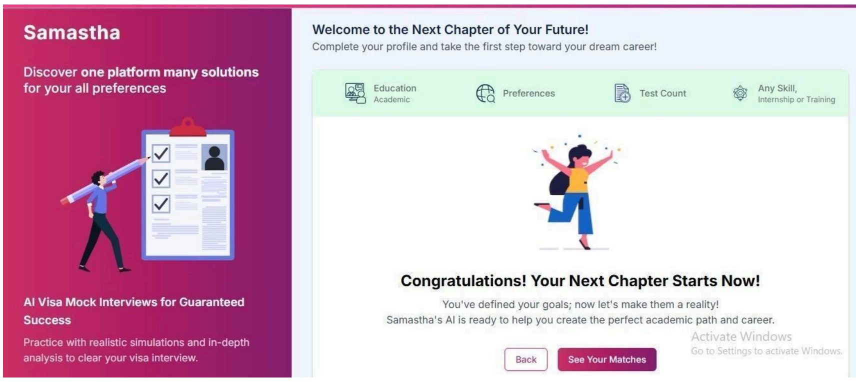

Figure 5.2: Profilecheck

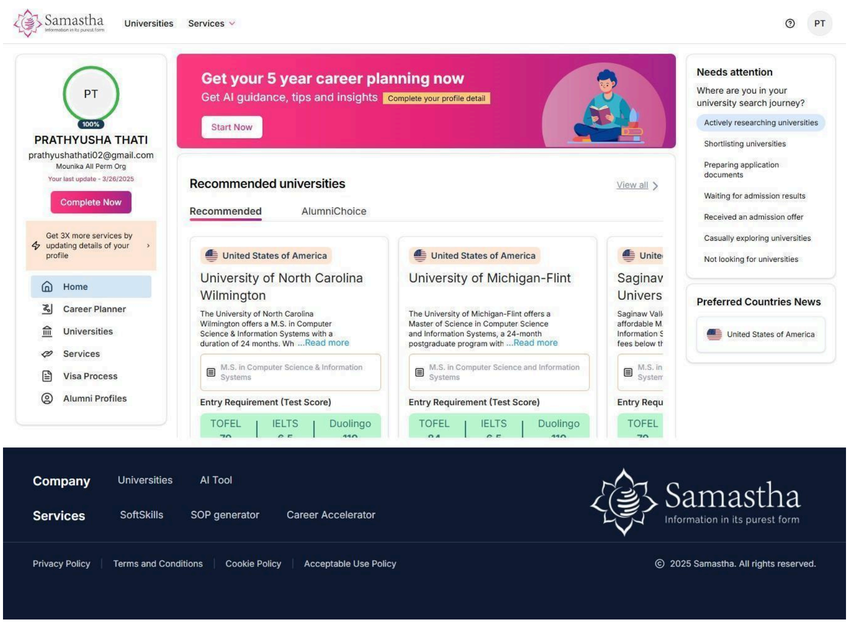

Figure 5.3: Test Page

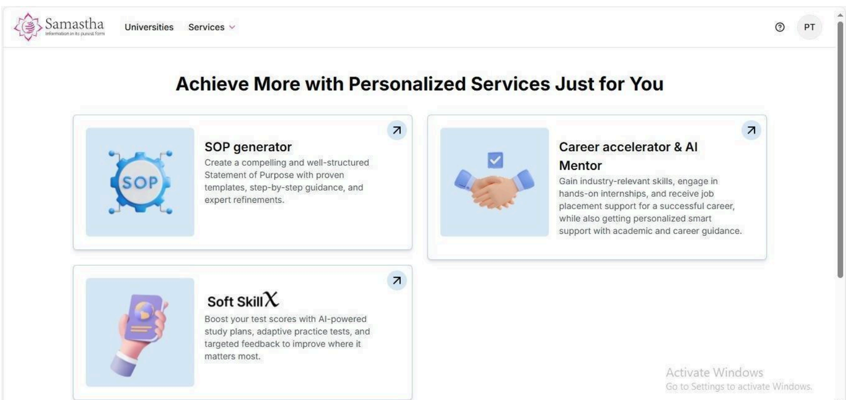

Figure 5.4: Services Page

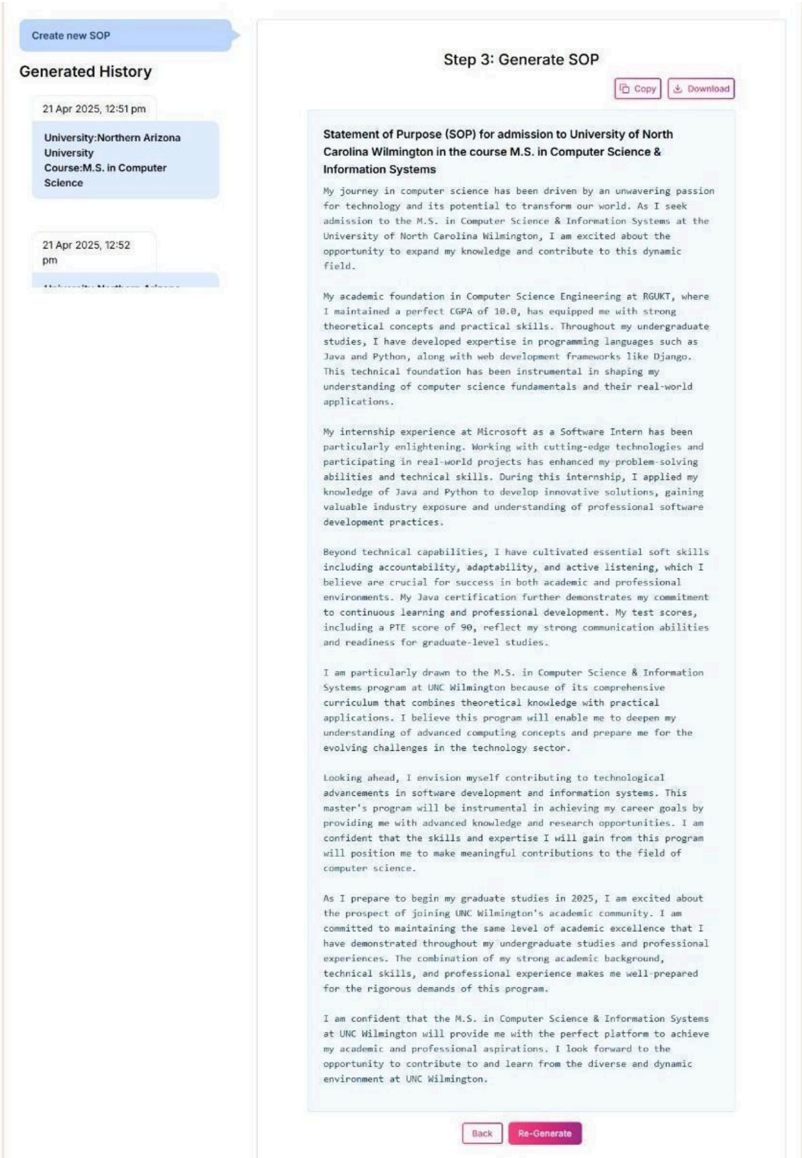

Figure 5.5: SOP Page

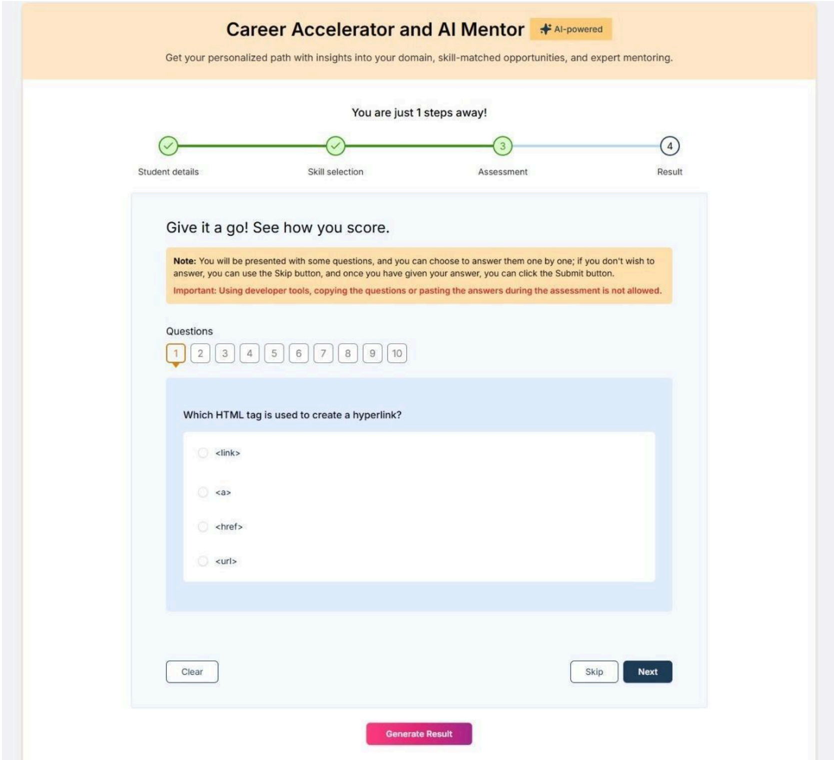

Figure 5.6: Career Accelerator Page

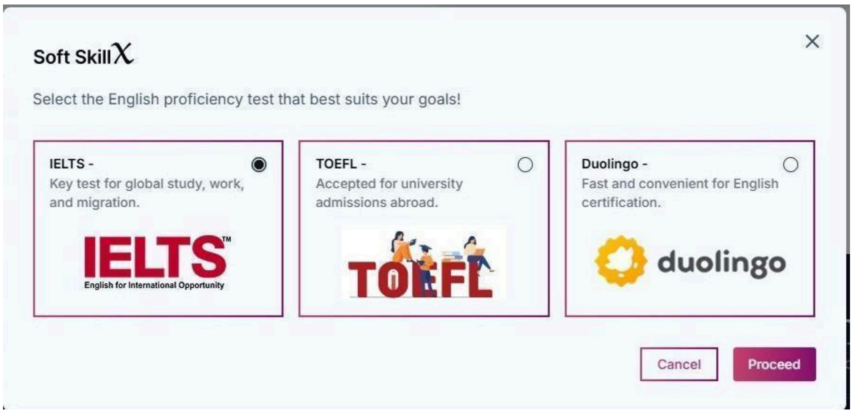

Figure 5.7: Soft SkillX

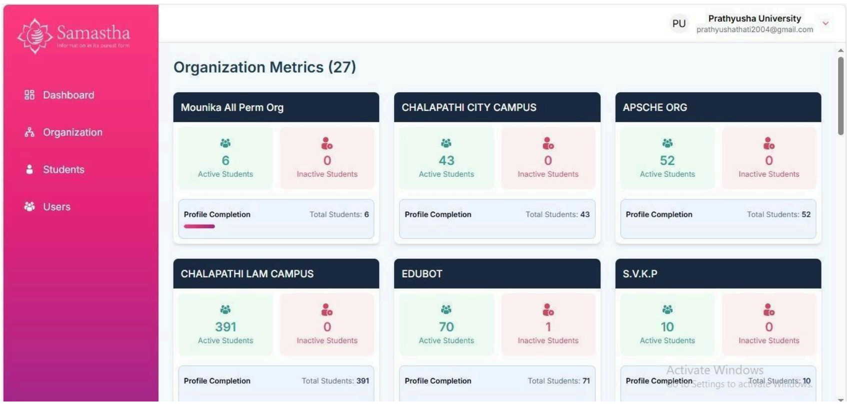

Figure 5.8: Organization Dashboard

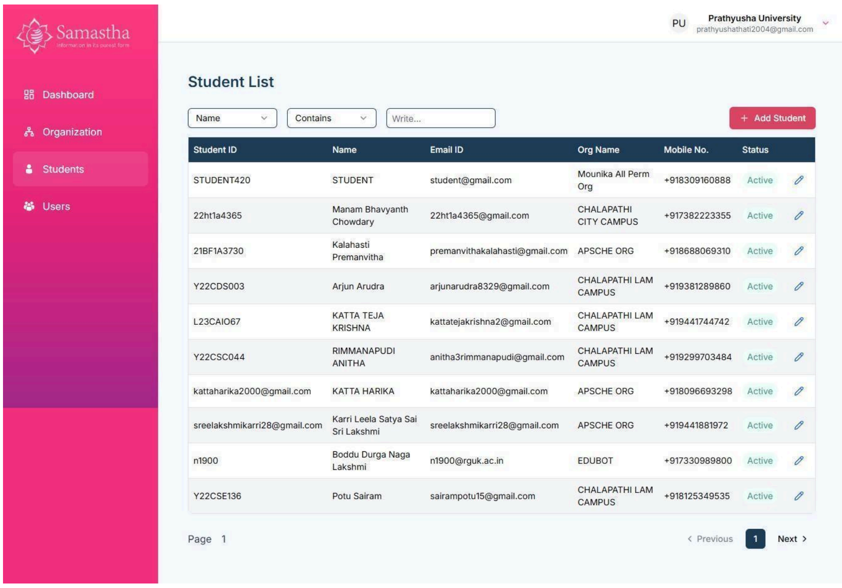

Figure 5.9: Student List Page

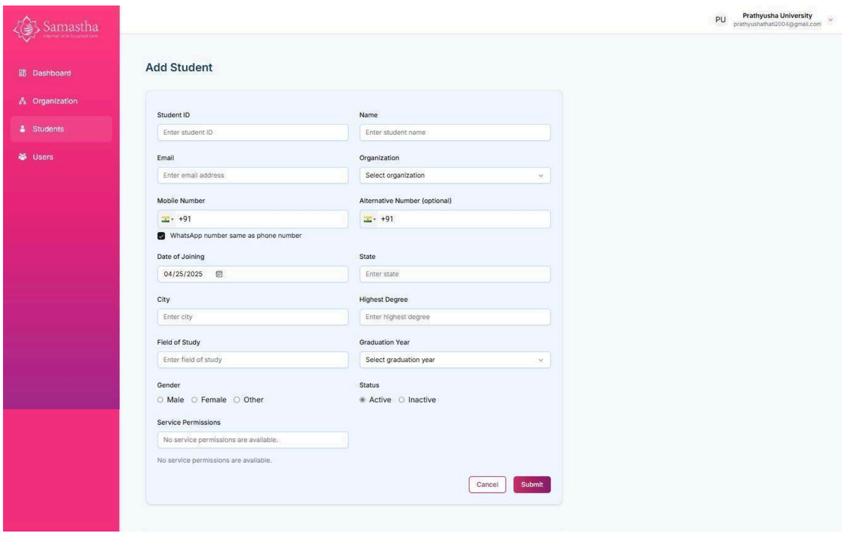

Figure 5.10: Student Add Page

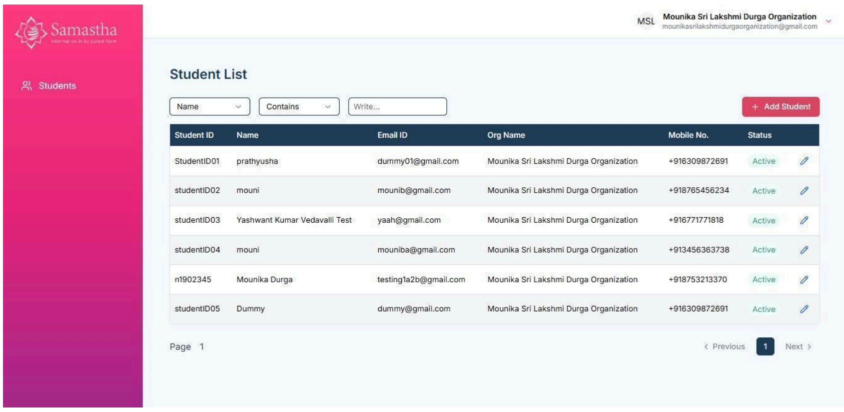

Figure 5.11: admin Page

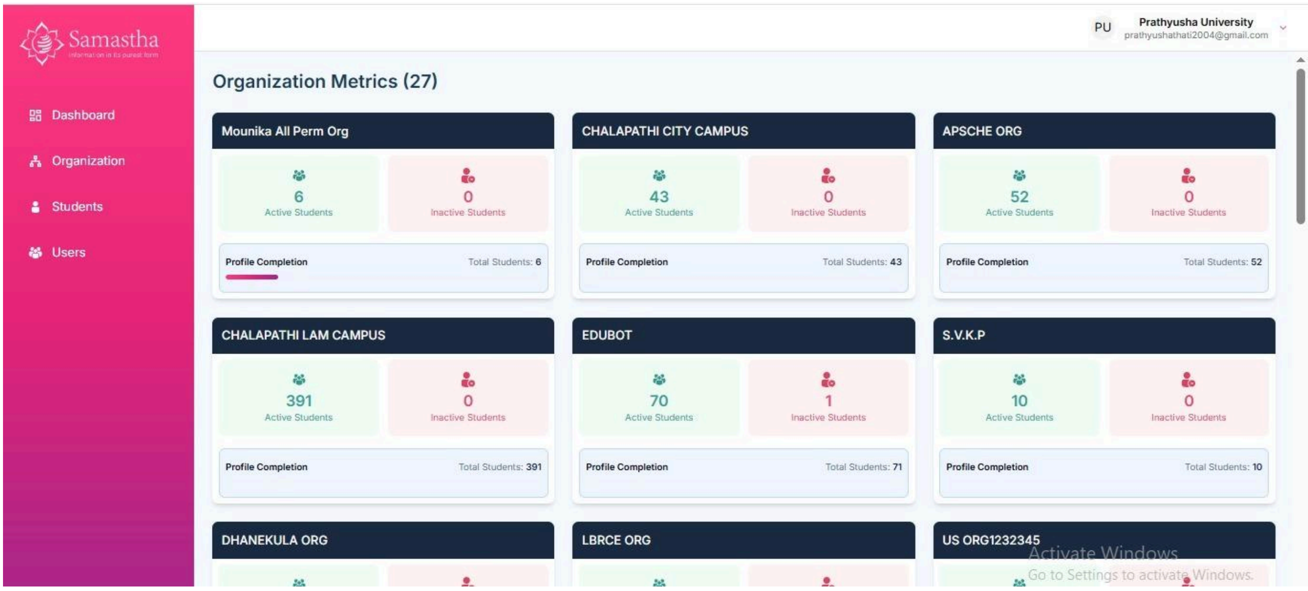

Figure 5.12: Organization Dashboard

0.6 Conclusion

Samastha provides a secure and user-friendly login experience for students seeking educational and career services [1]. The platform enables authentication through both email and phone number, verifying user identity via OTP [3]. Depending on the permissions associated with each user, Samastha intelligently redirects them to personalized service pages such as Career Accelerator, Soft SkillX, or Abroad Guidance [2]. This approach ensures a smooth transition into the platform's features while maintaining high standards of security and accessibility [6].

Samastha's structured login process reflects its commitment to offering a seamless and guided journey for students exploring international opportunities [1]. By combining robust authentication protocols with intuitive navigation [4], the platform minimizes entry barriers and enhances user trust from the very first interaction [7]. Furthermore, the adaptive redirection ensures that students are instantly connected with services most relevant to their goals, making their engagement more purposeful and efficient [2].

As students continue their journey within the platform, this strong foundation not only streamlines access but also sets the tone for a personalized and enriching experience [10]. Samastha's login system is more than just a gateway—it is the first step in a thoughtfully designed ecosystem that empowers students to make confident decisions about their educational and professional futures [1].

0.7 Future Scope / Directions

Samastha's future development roadmap focuses on enhancing both user experience and system intelligence through strategic technological integrations. The platform will introduce AI-driven personalized dashboards that dynamically adapt to individual user behaviors and academic needs, while improved accessibility features including advanced screen reader support and multilingual capabilities will ensure broader inclusivity [4]. Administrative functions will be strengthened through real-time analytics dashboards offering comprehensive insights into system usage and security metrics [7]. The authentication process will be streamlined via Single Sign-On (SSO) integration with existing institutional systems like learning management platforms [1], complemented by intelligent dashboards that automatically display role-specific services and shortcuts based on user permissions [2][7].

Security and support features will see significant upgrades, including an AI-powered monitoring system that analyzes login patterns and OTP delivery statistics to detect anomalies and provide contextual security recommendations [3]. The platform will implement smarter error messaging that delivers specific, actionable guidance rather than generic notifications [6]. Additionally, an AI virtual assistant with natural language processing capabilities will be introduced to handle user inquiries efficiently [5], alongside an academic counseling module that provides personalized career guidance by analyzing educational trends and institutional data [8]. These planned enhancements aim to create a more intuitive, secure, and responsive platform while maintaining the system's robust security framework.

0.8 References

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